

WEEK 4 – DATA ANALYTICS CAPSTONE PROJECT

Solved Business Analysis Report | Excel + Power BI Ready

Scope

EDA, sales regression, Power BI dashboard planning, SQL-style analysis results, findings and recommendations based on the supplied dataset.

Key KPIs

KPI	Value
Records	700
Original missing cells	53
Units Sold	1,125,806.0
Gross Sales	127,931,598.50
Net Sales	118,726,350.26
COGS	101,832,648.00
Profit	16,893,702.26
Profit Margin	14.23%

Main Findings

- Government has the highest total profit; Enterprise has negative aggregate profit.
- Paseo has the highest total net sales.
- France has the highest total profit among countries.
- The Low discount band has the highest aggregate profit and about 17.87% margin.
- 2014 has higher total net sales than 2013.

Regression

Net Sales predicted from Units Sold, Sale Price and Discounts. Test $R^2 = 0.7698$; MAE = 78,476.14; RMSE = 112,918.07.

Power BI Plan

- KPI cards
- Monthly Net Sales/Profit trend
- Product, Country and Segment charts
- Discount Band analysis
- Slicers and drill-down detail table

Recommendations

- Track margin with sales.
- Investigate Enterprise price, COGS and discount patterns.
- Evaluate discount depth.

- Use mix analysis before reallocating resources.
- Review monthly seasonality before setting targets.